Lindsey’s Technical Projects

Autonomous Underwater Vehicle

Objective:

- Use autonomous underwater vehicle to collect pH, light, and pressure data to assess the ability of aquatic plants to photosynthesize

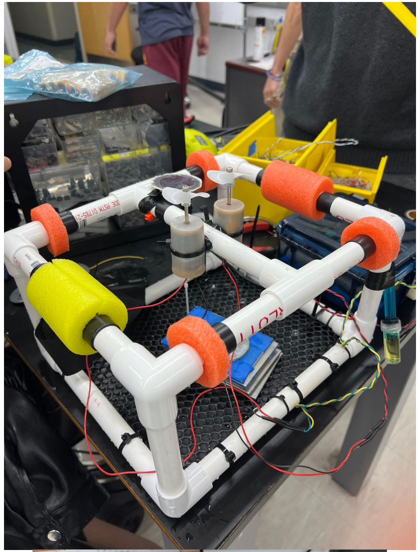
Overall Role:

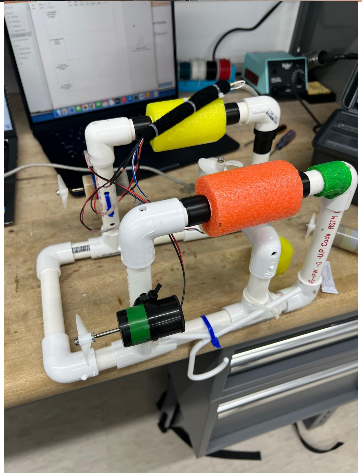
- Member of team of 4
- Lead team meetings
- Handled all CAD and 3D printing

Autonomous Underwater Vehicle

CAD and Design:

- Light sensor attachment for filter offset
- Oversized penetrator bolt

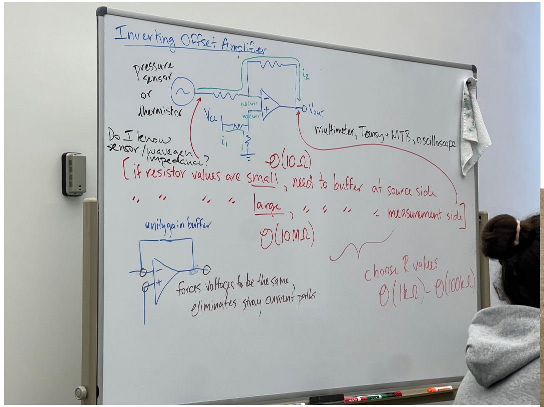


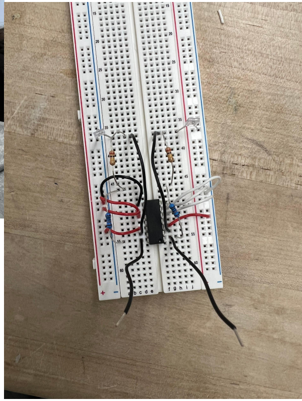


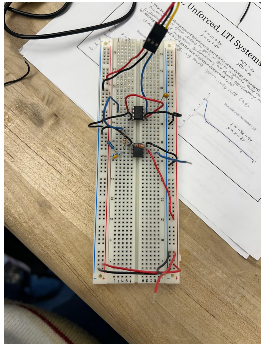
Autonomous Underwater Vehicle (continued)

Circuits:

- Designed and built op amp circuits to adjust voltage inputs from sensors to acceptable ranges
- Used oscilloscopes to observe inputs and outputs

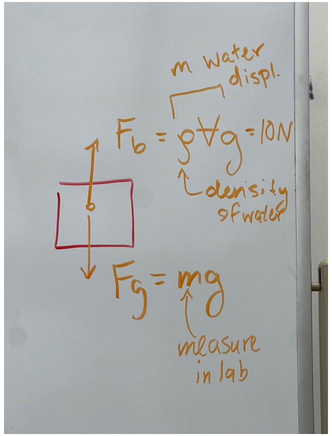






Autonomous Underwater Vehicle (continued)

- Conducted field tests with team
- Had problems with leaks but managed to collect data
- Calculated ballasts needed to offset buoyant force



Hammer

- Followed process routers for hard ace, soft face, handle, and head
- Used bandsaws, mills, lathes, and other shop machines
- Laser cut hammer stand
- Hand-sanded handle

French Horn Mouthpiece

- Modeled French horn mouthpiece in Solidworks
- 3D printed various iterations of the CAD model to fix issues
- Began to develop CAM pathway for machining on a CNC lathe

Lift Mate - Vacuum Pump Lift

- Worked on a team of 5 to design mechanic solution to lift vacuum pump
- designed parts in CAD
- Led team organizational efforts